

IP Addressing and Subnetting -- Guided Notes ANSWER KEY

Unit 2.3.1 | CompTIA Network+ N10-009 Obj. 2.3 | N10-008 2.3

For instructor use / distribute after students complete the notes.

Question 1

An IPv4 address is a _____ -bit number divided into four _____ -bit sections called octets. Each octet is written in _____ notation, with values ranging from 0 to _____.

Answer: 32-bit; 8-bit; dotted-decimal; 255.

Question 2

Convert the decimal number 172 to binary using the bit-position table (128, 64, 32, 16, 8, 4, 2, 1). Show your work step by step.

Answer: 128 fits (172 - 128 = 44). 64 doesn't fit. 32 fits (44 - 32 = 12). 16 doesn't fit. 8 fits (12 - 8 = 4). 4 fits (4 - 4 = 0). 2 and 1 don't fit. Result: 10101100.

Question 3

A subnet mask of 255.255.255.0 written in CIDR notation is _____. The mask tells a device which portion of an IP address is the _____ address and which portion identifies the _____.

Answer: /24; network; host.

Question 4

For the network 192.168.10.0/24, identify: the network address _____, the broadcast address _____, the number of usable host addresses _____, and the valid host range _____.

Answer: Network: 192.168.10.0; Broadcast: 192.168.10.255; Usable hosts: 254; Valid range: 192.168.10.1 to 192.168.10.254.

Question 5

A /26 subnet mask borrows _____ bits from the host portion of a /24. Each /26 subnet provides _____ usable host addresses. How many /26 subnets can be created from a single /24 network?

Answer: 2 bits; 62 usable hosts; 4 subnets (because $2^2 = 4$).

Question 6

VLSM stands for _____. Unlike fixed-length subnetting, VLSM allows a network engineer to assign _____ -sized subnets to different segments, which prevents _____ of IP address space.

Answer: Variable Length Subnet Masking; different (variable); waste.

Question 7

Explain the difference between a network address and a broadcast address. Why can neither be assigned to a host? What happens if a packet is sent to the broadcast address?

Answer: The network address (all host bits = 0) identifies the subnet itself; the broadcast address (all host bits = 1) represents all hosts on that subnet. Neither can be assigned to a host because they serve as reserved identifiers. A packet sent to the broadcast address is delivered to every device on that subnet.

Question 8 -- Real World Application

REAL WORLD: A network admin needs to allocate subnets for four departments: Engineering (50 hosts), Sales (25 hosts), HR (10 hosts), and a WAN link (2 hosts). Using VLSM starting from 10.0.0.0/24, identify the correct subnet size (/prefix) for each department and explain why you chose it.

Answer: Engineering: /26 (62 usable -- smallest mask that fits 50). Sales: /27 (30 usable -- fits 25). HR: /28 (14 usable -- fits 10). WAN link: /30 (2 usable -- exact fit for point-to-point). VLSM assigns the tightest mask that accommodates each group, conserving address space.